OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT

Electrical Load Estimation Report

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PRELIMINARY ILLUSTRATIVE REPORT. NOT FOR CONSTRUCTION. This report is an educational and preliminary engineering reference. It is not construction-ready, not code-compliant, not certified and not independently approved. Every input is an illustrative assumption for the sample project, chosen for ideal conditions. It requires review by suitably qualified engineers before any use beyond education.

1. Executive summary

The load model gives a facility demand of 2,697.3 kW (2,865.6 kVA at a power factor of 0.941) for an IT critical load of 2,000.0 kW. The design-point PUE is 1.349 against a target of 1.35. In normal operation each path carries 1,348.7 kW and 1,348.7 kW; if one path is lost, the other carries all 2,697.3 kW (2,865.6 kVA). Checks: PASS 9, WARNING 2, NOT VERIFIED 25. A PASS is a software check against the stated assumptions, not an engineering approval.

2. Purpose and scope

This report estimates the connected and demand electrical load of the reference data centre, the UPS and distribution losses, the design-point PUE and the allocation of load to the A and B power paths. It does not size transformers, generators, UPS or batteries (Stage 3), cables (Stage 4), or perform short-circuit, protection, earthing or power-quality studies (Stages 4 and 5).

3. Design basis

Parameter	Value	Unit	Status
IT critical load	2000	kW	ASSUMPTION
Target PUE	1.35	-	ASSUMPTION
Initial estimated total facility demand	2700	kW	ASSUMPTION
Initial design power factor	0.95	-	ASSUMPTION
Future expansion allowance	0	%	ASSUMPTION
Incoming MV supply voltage	13.8	kV	ASSUMPTION
LV line-to-line voltage	400	V	ASSUMPTION
System frequency	60	Hz	ASSUMPTION
Power architecture	Dual independent A and B power paths	-	ASSUMPTION
UPS topology	2N	-	ASSUMPTION
Generator concept	N+1 per power path	-	ASSUMPTION

4. Input data and sources

The load schedule below is read from config/load_schedule.yaml. Every item is an illustrative assumption for the sample project, not a measured or manufacturer value.

ID	Description	Qty x kW	PF	DF	Path A	Status
IT-1	IT equipment, all halls (dual-corded)	1 x 2000	0.95	1.00	0.50	ASSUMPTION
MC-1	Chillers (3 of 4 run)	4 x 120	0.90	0.75	0.50	ASSUMPTION
MC-2	CRAH fan units	20 x 6	0.95	1.00	0.50	ASSUMPTION

ID	Description	Qty x kW	PF	DF	Path A	Status
MC-3	Chilled-water and condenser pumps (4 of 6 run)	6 x 15	0.85	0.67	0.50	ASSUMPTION
LS-1	Lighting	1 x 15	0.90	0.80	0.50	ASSUMPTION
LS-2	Small power and maintenance sockets	1 x 20	0.90	0.50	0.50	ASSUMPTION
AUX-1	Fire, security, BMS and EPMS	1 x 25	0.90	1.00	0.50	ASSUMPTION

5. Assumptions

All of the following are assumptions made for the sample project. They are not manufacturer data. They are listed in config/assumptions_register.yaml with their confirmation status (UNCONFIRMED).

Assumption	Value	Unit
UPS efficiency	96	%
Distribution losses	1.0	%
Future expansion allowance	0	%
Initial design power factor	0.95	-
Target PUE	1.35	-
Operating arrangement	Chillers 3 of 4 run, pumps 4 of 6 run, all IT dual-corded and split 50/50 between paths	-
Path-out scenarios	All load transfers to the surviving path	-

6. Calculation methodology

For each item, connected load = quantity x unit rating, and demand = connected load x demand factor. Reactive power = demand kW x tan(acos(power factor)). Totals add real and reactive power separately and apparent power is sqrt(P^2 + Q^2). UPS losses = IT demand x (1 / efficiency - 1). Distribution losses = loss percentage x (load items + UPS losses). Facility demand = load items + UPS losses + distribution losses. PUE = facility kW / IT kW at the design point. Path A load sums each item's demand times its path A fraction, plus the UPS losses by the IT split, plus its own distribution losses; path B is the remainder.

PUE definition and limits: the figure here is a design-point power ratio. The metric as usually reported is an energy ratio over a period, which depends on IT utilisation, climate and operating modes that this model does not simulate.

7. Results

Category	Demand (kW)
it_critical	2,000.00
mechanical_cooling	540.30
lighting_small_power	22.00
auxiliary	25.00

Quantity	Value	Unit
Connected load of all items	2,750.0	kW
Demand of all load items	2,587.30	kW
UPS losses	83.33	kW
Distribution losses	26.71	kW
Facility demand	2,697.34	kW
Facility reactive demand	967.47	kvar
Facility apparent demand	2,865.60	kVA
Facility power factor	0.9413	-
Design-point PUE	1.3487	-
Design demand including growth allowance	2,697.34	kW

Quantity	Value	Unit
Design apparent demand including growth	2,865.60	kVA

A/B allocation	kW	kvar	kVA
Normal: path A	1,348.67	483.73	1,432.80
Normal: path B	1,348.67	483.73	1,432.80
One path out: surviving path carries all	2,697.34	967.47	2,865.60

8. Design checks and status

Statuses: PASS (the check ran and found no problem, not an approval), WARNING (a possible inconsistency), FAIL (a defect that blocks release), NOT VERIFIED (cannot be judged without data or a limit that has not been supplied), NOT APPLICABLE.

Check	Status	Detail
Inputs needed up to Stage 5 are supplied	PASS	All required user inputs for the current stage are present.
Manufacturer data needed up to Stage 5 is supplied	NOT VERIFIED	18 manufacturer data item(s) pending: equipment.generator_esp_rating, equipment.generator_cop_rating, generators.prp_max_average_load_factor_pct, generators.operating_hours_restriction, generators.overload_capability, generators.max_step_load_pct, generators.max_frequency_dip_pct, generators.max_voltage_dip_pct, generators.ups_rectifier_walk_in_s, generators.ups_input_current_thd_pct, battery.dc_bus_voltage_v, battery.inverter_efficiency_pct, battery.end_of_life_capacity_factor, battery.temperature_correction_factor, battery.constant_power_discharge_data, short_circuit.transformer_x_r_ratio, short_circuit.generator_subtransient_reactance_pct, short_circuit.generator_x_r_ratio
Inputs needed in later stages	PASS	Nothing is deferred.
Facility demand equals IT load times target PUE	PASS	2000 kW x 1.35 = 2,700.0 kW against 2700 kW stated (0.00 % apart).
LV line and phase voltages are consistent	PASS	400 V / 230 V = 1.739 against sqrt(3) = 1.732.
LV voltage and frequency pairing	NOT VERIFIED	400/230 V at 60 Hz is an unusual pairing. The project owner's decision is recorded and has not been independently verified: Project owner selected 400/230 V at 60 Hz for this reference project and stated that the serving utility confirmed it. Not independently verified.
Equipment quantities divide evenly between power paths	PASS	Transformer and UPS quantities divide evenly between 2 paths.
Transformer secondary full-load current	NOT VERIFIED	3150 kVA at 400 V gives 4,547 A. Whether LV switchgear and busduct exist at this current, and the prospective fault level, must be verified in Stages 4 and 5. No limit is applied here.
IT load in the schedule matches the design basis	PASS	Connected IT load in the schedule 2,000.0 kW against the design basis 2000 kW.
Design-point PUE against the target	NOT VERIFIED	Design-point PUE 1.349 against a target of 1.35. A design-point power ratio under the stated assumptions, not an annual energy PUE. CALCULATION RESULT: PASS (preliminary). OVERALL STATUS: NOT VERIFIED, because 7 input(s) are not verified (worst: UNVERIFIED_PLACEHOLDER). A calculation on unverified inputs is not evidence that the design works.
Load model against the initial facility demand estimate	PASS	Load model 2,697.3 kW against the initial estimate 2700 kW (-0.10 %). The initial estimate is not adjusted to match. 2 % is a review trigger, not an engineering limit.
Facility power factor against the design assumption	PASS	Facility power factor 0.941 against the design assumption 0.95. 0.02 is a review trigger, not an engineering limit.
A/B path allocation adds up to the facility load	PASS	Path A 1,348.7 kW + path B 1,348.7 kW against 2,697.3 kW. A software consistency check, not an engineering verification.

Check	Status	Detail
Transformer loading, worst credible contingency	WARNING	Worst credible case TX-PATH-OUT: 2,866 kVA on 1 x 3,150 kVA derated = 3,150 kVA, loading 91.0 % of the site-derated continuous rating (target 90 %, maximum 100 %), within the maximum. Against the nameplate rating it is 91.0 %, shown for information only: the status uses the derated rating. Preliminary: derating factors are placeholders, no overload used. Rating per unit to meet the target 3,184 kVA (suggested size 4,000). ACCEPTED DESIGN-TARGET EXCEPTION EX-001 (conditionally accepted by Project owner, 2026-11-05): the result is within the accepted ceiling of 91 %, pending independent technical verification. The check stays a WARNING. This is a preliminary result on unverified inputs (worst: UNVERIFIED_PLACEHOLDER).
UPS loading in kW, worst credible contingency	NOT VERIFIED	Worst credible case UPS-PATH-OUT: 2,000 kW on 1 x 2,500 kW derated = 2,500 kW, loading 80.0 % of the site-derated continuous rating (target 80 %, maximum 100 %), within the maximum. Against the nameplate rating it is 80.0 %, shown for information only: the status uses the derated rating. Preliminary: derating factors are placeholders, no overload used. Rating per unit to meet the target 2,500 kW (suggested size 2,500). CALCULATION RESULT: PASS (preliminary). OVERALL STATUS: NOT VERIFIED, because 11 input(s) are not verified (worst: UNVERIFIED_PLACEHOLDER). A calculation on unverified inputs is not evidence that the design works.
UPS loading in kVA, worst credible contingency	WARNING	Worst credible case UPS-PATH-OUT: 2,105 kVA on 1 x 2,500 kVA derated = 2,500 kVA, loading 84.2 % of the site-derated continuous rating (target 80 %, maximum 100 %), within the maximum. Against the nameplate rating it is 84.2 %, shown for information only: the status uses the derated rating. Preliminary: derating factors are placeholders, no overload used. Rating per unit to meet the target 2,632 kVA (suggested size 3,000). This is a preliminary result on unverified inputs (worst: UNVERIFIED_PLACEHOLDER).
Generator loading in kVA, worst credible N+1 contingency	NOT VERIFIED	Worst credible case GEN-PATH-OUT: 2,854 kVA on 3 x 2,000 kVA derated = 6,000 kVA, loading 47.6 % of the site-derated continuous rating (target 80 %, maximum 100 %), within the maximum. Against the nameplate rating it is 47.6 %, shown for information only: the status uses the derated rating. Preliminary: derating factors are placeholders, no overload used. Rating per unit to meet the target 1,189 kVA (suggested size 1,250). CALCULATION RESULT: PASS (preliminary). OVERALL STATUS: NOT VERIFIED, because 11 input(s) are not verified (worst: UNVERIFIED_PLACEHOLDER). A calculation on unverified inputs is not evidence that the design works.
Generator loading in kW, worst credible N+1 contingency	NOT VERIFIED	Worst credible case GEN-PATH-OUT: 2,687 kW on 3 x 1,600 kW derated = 4,800 kW, loading 56.0 % of the site-derated continuous rating (target 80 %, maximum 100 %), within the maximum. Against the nameplate rating it is 56.0 %, shown for information only: the status uses the derated rating. Preliminary: derating factors are placeholders, no overload used. Rating per unit to meet the target 1,120 kW. CALCULATION RESULT: PASS (preliminary). OVERALL STATUS: NOT VERIFIED, because 12 input(s) are not verified (worst: UNVERIFIED_PLACEHOLDER). A calculation on unverified inputs is not evidence that the design works.
Generator rating basis confirmed	NOT VERIFIED	The 2000 kVA rating is ASSUMED to be a PRP rating. Manufacturer documentation has not confirmed it, and it must not be treated as one until it does. The rated power factor (0.8) is a placeholder.
Generator load steps and transient performance	NOT VERIFIED	Upper-bound steps: STEP-1 2,215 kVA (36.9 % of running capacity), STEP-2 601 kVA (10.0 % of running capacity), STEP-3 41 kVA (0.7 % of running capacity). Upper bound: each block applied at once at its steady-state demand. Soft starting, rectifier walk-in and sequencing are not modelled. Transient performance is not verified: manufacturer step, dip, walk-in and harmonic data are required.
Generator PRP load-factor and operating-hour limits	NOT VERIFIED	Worst credible loading 47.6 % is an upper bound on the average load factor during an outage. The manufacturer's average load factor limit and operating-hour restriction for the PRP rating are not yet supplied.
Battery sizing	NOT VERIFIED	2,000 kW for 10 min is 333 kWh at the UPS output (arithmetic, not a battery size). No battery is sized without the manufacturer's constant-power discharge data. Missing: battery.dc_bus_voltage_v, battery.inverter_efficiency_pct, battery.end_of_life_capacity_factor, battery.temperature_correction_factor, battery.design_margin_factor, battery.constant_power_discharge_data.

Check	Status	Detail
Derating factors	NOT VERIFIED	10 derating factor(s) are placeholders of 1.00 for ideal conditions (30 degC, 0 m): no derating is applied. Replace them with the manufacturer's values at the site conditions: derating.transformer_ambient_factor, derating.transformer_altitude_factor, derating.transformer_installation_factor, derating.ups_ambient_factor, derating.ups_altitude_factor, derating.ups_installation_factor, derating.generator_ambient_factor, derating.generator_altitude_factor, derating.generator_installation_factor, derating.transformer_contingency_factor
Cable voltage drop	NOT VERIFIED	CAB-MV-A: 120 A, drop 0.05 %; CAB-GEN-A: 2,887 A on 6 runs (481 A per run), drop 0.47 %; CAB-UPS-IN-A: 3,037 A on 6 runs (506 A per run), drop 0.20 %; CAB-UPS-OUT-A: 3,039 A on 6 runs (506 A per run), drop 0.25 %; CAB-MECH-A: 860 A on 3 runs (287 A per run), drop 0.50 %. Limit 3 % (a placeholder criterion). Resistance is derived at the stated conductor temperature; reactance is a placeholder. CALCULATION RESULT: PASS (preliminary). OVERALL STATUS: NOT VERIFIED, because 18 input(s) are not verified (worst: UNVERIFIED_PLACEHOLDER). A calculation on unverified inputs is not evidence that the design works.
Cable current rating against the design current	NOT VERIFIED	CAB-MV-A: needs 120 A per run after derating; CAB-GEN-A: needs 481 A per run after derating; CAB-UPS-IN-A: needs 506 A per run after derating; CAB-UPS-OUT-A: needs 506 A per run after derating; CAB-MECH-A: needs 287 A per run after derating. No tabulated rating has been supplied, so none is assumed and the cables cannot be judged. Compare each required rating with the cable table or data sheet for the installation method.
Busduct rating against the design current	NOT VERIFIED	BD-TX-A: 4,136 A, needs 4,136 A rated after derating. The busduct rating is manufacturer data and is not supplied.
Busduct voltage drop	NOT VERIFIED	Busduct impedance is manufacturer data and is not supplied, so the voltage drop cannot be calculated.
Switchgear rated current and verification data	NOT VERIFIED	LV-MSB-A: busbars need 4,136 A rated (missing: rated_current_a, short_circuit_rating_ka, form_of_separation, ip_rating, verification_evidence); MV-SWG-A: busbars need 120 A rated (missing: rated_current_a, short_circuit_rating_ka, form_of_separation, ip_rating, verification_evidence). Ratings, separation, IP rating and IEC 61439 or IEC 62271 verification evidence are manufacturer data and are not supplied.
Distribution selections are placeholders	NOT VERIFIED	Cable sizes, runs and lengths in config/distribution.yaml are illustrative placeholders. Only path A is modelled; path B is assumed identical by symmetry. A selection is not a verified design.
Prospective short-circuit levels	NOT VERIFIED	LV bus calculated: Transformer-limited upper bound 82.6 kA. NOT CALCULATED, MV bus, utility: missing short_circuit.utility_fault_level_mva, short_circuit.utility_x_r_ratio. NOT CALCULATED, LV bus, utility and one transformer: missing short_circuit.transformer_x_r_ratio, short_circuit.utility_fault_level_mva, short_circuit.utility_x_r_ratio. NOT CALCULATED, LV bus, generators: missing short_circuit.generator_subtransient_reactance_pct, short_circuit.generator_x_r_ratio. Engineering impact and the documents needed: Generator subtransient reactance x''d (short_circuit.generator_subtransient_reactance_pct): Sets the generator fault contribution. Without it the generator case is NOT CALCULATED. Source needed: Generator set data sheet: subtransient reactance x''d, X/R, rated power factor. Generator X/R ratio (short_circuit.generator_x_r_ratio): Sets the peak current from generators. Source needed: Generator set data sheet: subtransient reactance x''d, X/R, rated power factor. Transformer X/R ratio (short_circuit.transformer_x_r_ratio): Sets the peak factor on the LV side. Source needed: Transformer test report (short-circuit impedance and load losses). Utility initial short-circuit power at the MV connection (short_circuit.utility_fault_level_mva): Sets every fault level in the facility. Without it no MV or LV fault level can be calculated. Source needed: Utility written statement of maximum and minimum fault levels and X/R at the point of connection. Utility X/R ratio (short_circuit.utility_x_r_ratio): Sets the peak factor and the DC component of the fault current. Source needed: Utility written statement of maximum and minimum fault levels and X/R at the point of connection. Maximum three-phase level at the busbars, IEC 60909 method. Cable and busduct impedance between the transformer and the bus are ignored, which is conservative. Motor contribution (the mechanical plant motors) is not included. It adds to the LV level and needs motor data. Minimum fault current, needed for protection sensitivity, is not calculated. A result is NOT VERIFIED until the utility, transformer and generator data are confirmed.

Check	Status	Detail
LV fault level against the practical limit	NOT VERIFIED	Only the transformer-limited upper bound can be calculated: 82.6 kA, valid for any utility fault level. It does not exceed the 100 kA practical limit (Ohmetry preliminary policy). The design level needs the utility and generator data.
Short-circuit withstand of cables, busducts and switchgear	NOT VERIFIED	BD-TX-A: required level not calculable; LV-MSB-A: required level not calculable; MV-SWG-A: required level not calculable. A rating that is not supplied is not assumed, so the item cannot be judged. A peak withstand above the peak current is needed as well.
Cable thermal withstand against the fault level	NOT VERIFIED	The fault current is not available, so no cable can be judged. Adiabatic equation with k = 143 (reference value, verify against the cable data sheet) and an assumed clearing time of 0.5 s, a placeholder until the protection study. Fault taken at the bus the cable leaves, split equally over parallel runs.
Protection devices: breaking capacity and settings	NOT VERIFIED	MV-CB-A: required breaking capacity not calculable; ACB-TX-A: required breaking capacity not calculable; ACB-GEN-A: required breaking capacity not calculable; ACB-UPS-IN-A: required breaking capacity not calculable. Device ratings, time-current curves and settings are manufacturer data and are not supplied. Discrimination, arc-flash and sensitivity to minimum fault current are not assessed.
Earthing: grid resistance and ground potential rise screening	NOT VERIFIED	NOT CALCULATED. Missing: Soil resistivity (earthing.soil_resistivity): Sets the grid resistance and ground potential rise. Without it the earthing screening is NOT CALCULATED. Source needed: Site soil resistivity survey (Wenner method) report. Earth grid area (earthing_study.grid_area_m2): Sets the grid resistance. Source needed: Earthing grid layout drawing and design. Total buried grid conductor length (earthing_study.grid_conductor_length_m): Sets the grid resistance. Source needed: Earthing grid layout drawing and design. Grid burial depth (earthing_study.grid_burial_depth_m): Sets the grid resistance. Source needed: Earthing grid layout drawing and design. No default is substituted.
Manufacturer-data register is complete and honest	NOT VERIFIED	The register covers every manufacturer-data item and placeholder (83 entries). 83 are NOT VERIFIED: 59 pending, 0 placeholders, 0 supplied and awaiting review. Placeholders are not validated equipment data.

9. Limitations

The load list is illustrative. Demand factors depend on the real operating arrangement. UPS losses assume all IT load is on the UPS at one efficiency. Distribution losses use a single percentage. Power factors of losses use the design assumption. PUE is a design-point ratio, not an annual energy figure. Path-out scenarios assume all load transfers to the surviving path and ignore load shedding and equipment limits. No harmonic, step-loading, transient or short-circuit effects are included. A spreadsheet or script is not a substitute for specialist software or manufacturer sizing tools where those are required.

10. Outstanding information

The following inputs are needed in later stages and are not yet supplied: none.

11. Required engineering review

The design basis, every assumption in the register and every result require independent review by suitably qualified engineers. No review has taken place. Release gate: DEVELOPMENT.

12. Revision history

Revision	Date	Description	Reviewer
A5	2026-11-05	First generated load estimate (Stage 2). Illustrative assumptions for the sample project.	not assigned